

6 **(long)** A series of two-dimensional aerofoils are laid out in a vertical cascade, such that they deflect the upstream flow from an angle α_1 to an angle α_2 with the horizontal, as shown in Fig. 4. The figure portrays the flow around one of the aerofoils, with streamlines A and B parallel and separated by a vertical distance L . The flow upstream has uniform velocity U_∞ , while the flow downstream is non-uniform and has magnitude $V - \delta V \cos(2\pi y/L)$, where $y = \pm L/2$ are the vertical coordinates of streamlines A and B. The pressure upstream and downstream of the cascade is the ambient pressure.

- (a) Find the value of V in terms of U_∞ , α_1 and α_2 . [5]
- (b) The streamlines A and B can be considered inviscid. Find the value of δV in terms of U_∞ , α_1 and α_2 . [5]
- (c) Find the value and direction of the horizontal force per unit depth into the page, F_x , exerted by the flow on the aerofoil, as a function of ρ , L , U_∞ , α_1 and α_2 . Find the ratio of F_x to $1/2\rho U_\infty^2 L$ when $\alpha_1 = \arccos(3/5)$ and $\alpha_2 = \arccos(4/5)$. [10]
- (d) Find the value and direction of the vertical force per unit depth into the page, F_y , exerted by the flow on the aerofoil, as a function of ρ , L , U_∞ , α_1 and α_2 . Find the ratio of F_y to $1/2\rho U_\infty^2 L$ when $\alpha_1 = \arccos(3/5)$ and $\alpha_2 = \arccos(4/5)$. [10]

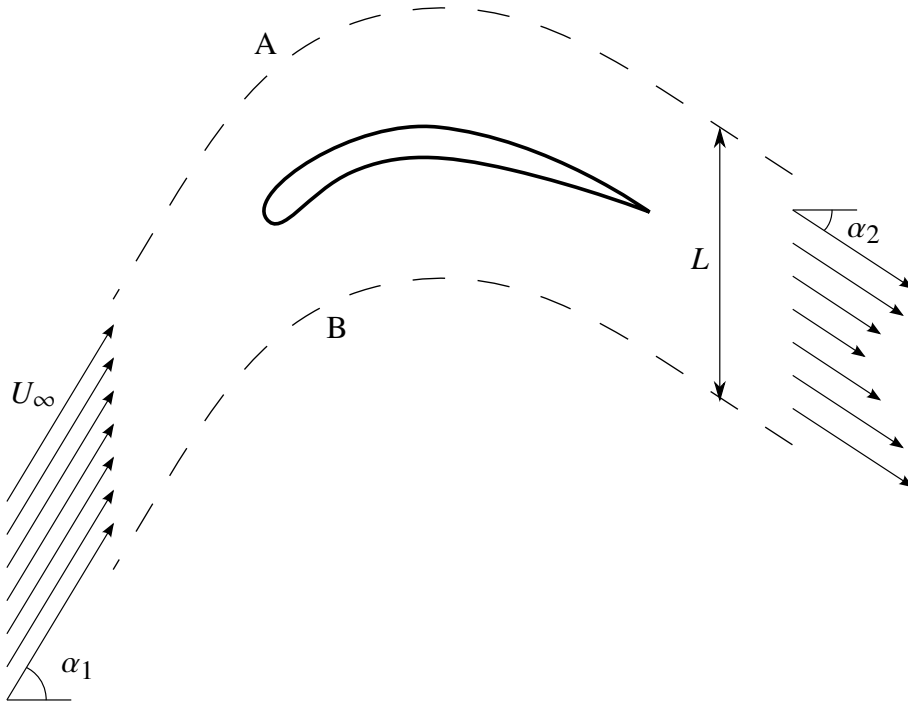


Fig. 4